

Definition: Given a set A , $\mathcal{P}(A) := \{\text{all subsets of } A\}$

Definition: We say $P \in \mathcal{P}(A)$ is a partition of A if

- $\bigcup_{X \in P} X = A$
- $X \neq Y \in P \Rightarrow X \cap Y = \emptyset$

example: $A = \{1, 2, 3, 4, 5, 6\}$,

$$P = \{\{1\}, \{2, 3, 4\}, \{5, 6\}\}$$

$$A = \mathbb{Z} \quad P = \left\{ \begin{array}{l} \{0, \pm 3, \pm 6, \dots\} \\ \{\dots, -2, 1, 4, 7, \dots\} \\ \{\dots, -1, 2, 5, 8, \dots\} \end{array} \right\}$$

Theorem: There is a one to one correspondence between partitions of A and equivalence relations on A .

proof: Suppose P is a partition of A . Define a relation $R \subseteq A \times A$ as $(a, b) \in R \iff a, b \in X \in P$
Check that this is an equivalence relation:

① $(a, a) \in R$, because $a \in X$ for some $X \in P$, can $x \cup X = A$

② Suppose $(a, b) \in R \Rightarrow a, b \in X \in P \Rightarrow b, a \in X \in P \Rightarrow (b, a) \in R$

③ Suppose $(a, b) \in R, (b, c) \in R$
 $\Rightarrow \left. \begin{array}{l} a, b \in X \in P \\ b, c \in Y \in P \end{array} \right\} b \in X \cap Y \Rightarrow X = Y$
 $\Rightarrow a, c \in X \Rightarrow (a, c) \in R$

Suppose $R \subseteq A \times A$ is an equivalence relation.

$$P := \{[a] \mid a \in A\} \quad ([a] = \{b \in A \mid (a, b) \in R\})$$

Suppose $a \in A \Rightarrow (a, a) \in R \Rightarrow a \in [a] = \bigcup_{[c] \in P} [c]$

$$\Rightarrow A \subseteq \bigcup_{[c] \in P} [c] \Rightarrow \bigcup_{[c] \in P} [c] = A$$

Take $a, b \in A$, consider $[a] \cap [b]$. Suppose $x \in [a] \cap [b]$
 $\Rightarrow x \in [a]$ and $x \in [b]$

$$(a, x) \in R \quad (b, x) \in R \Rightarrow (a, b) \in R$$

$$[a] = \{y \in A \mid (a, y) \in R\} = \{y \in A \mid (b, y) \in R\} = [b]. \quad \square$$